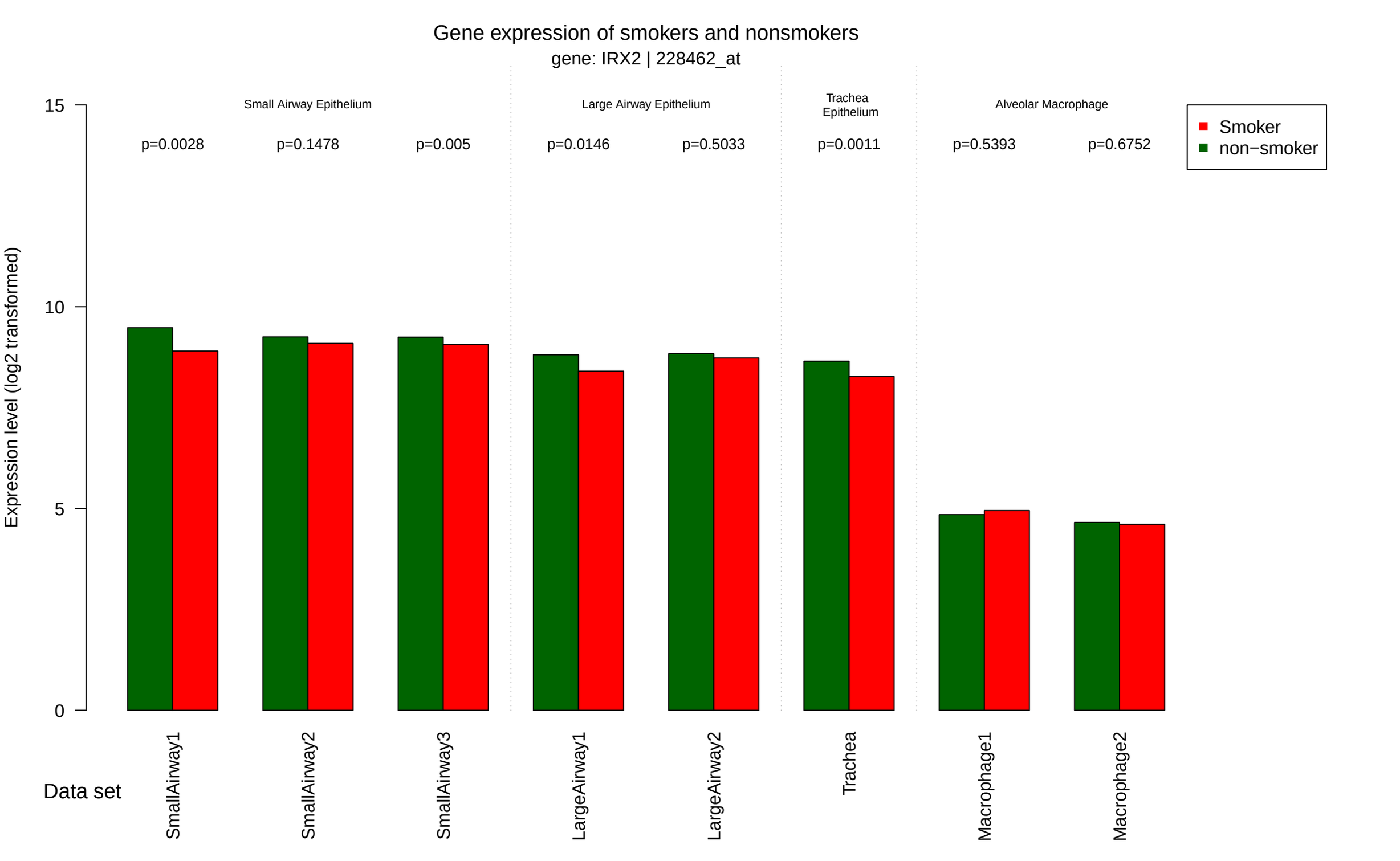
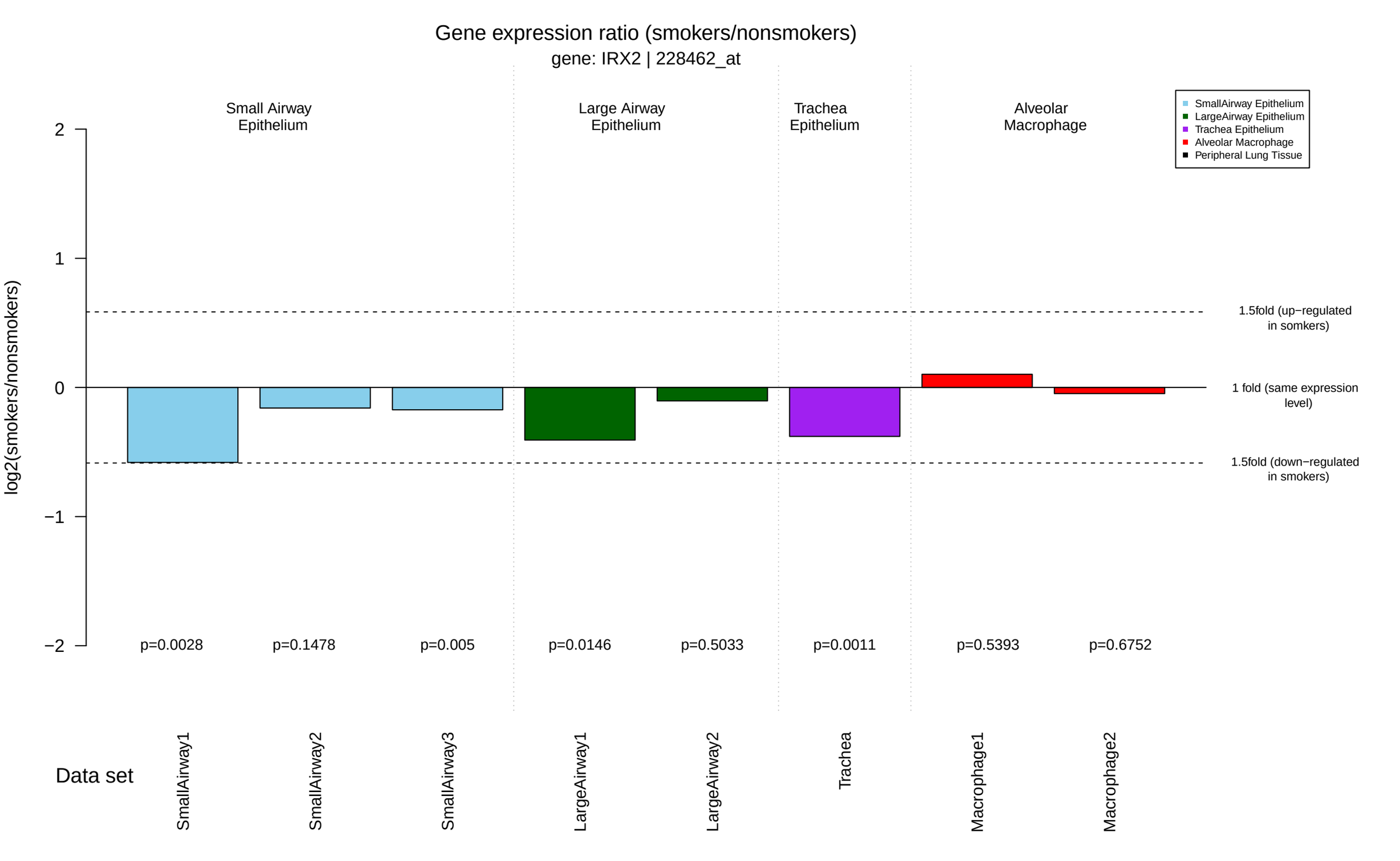
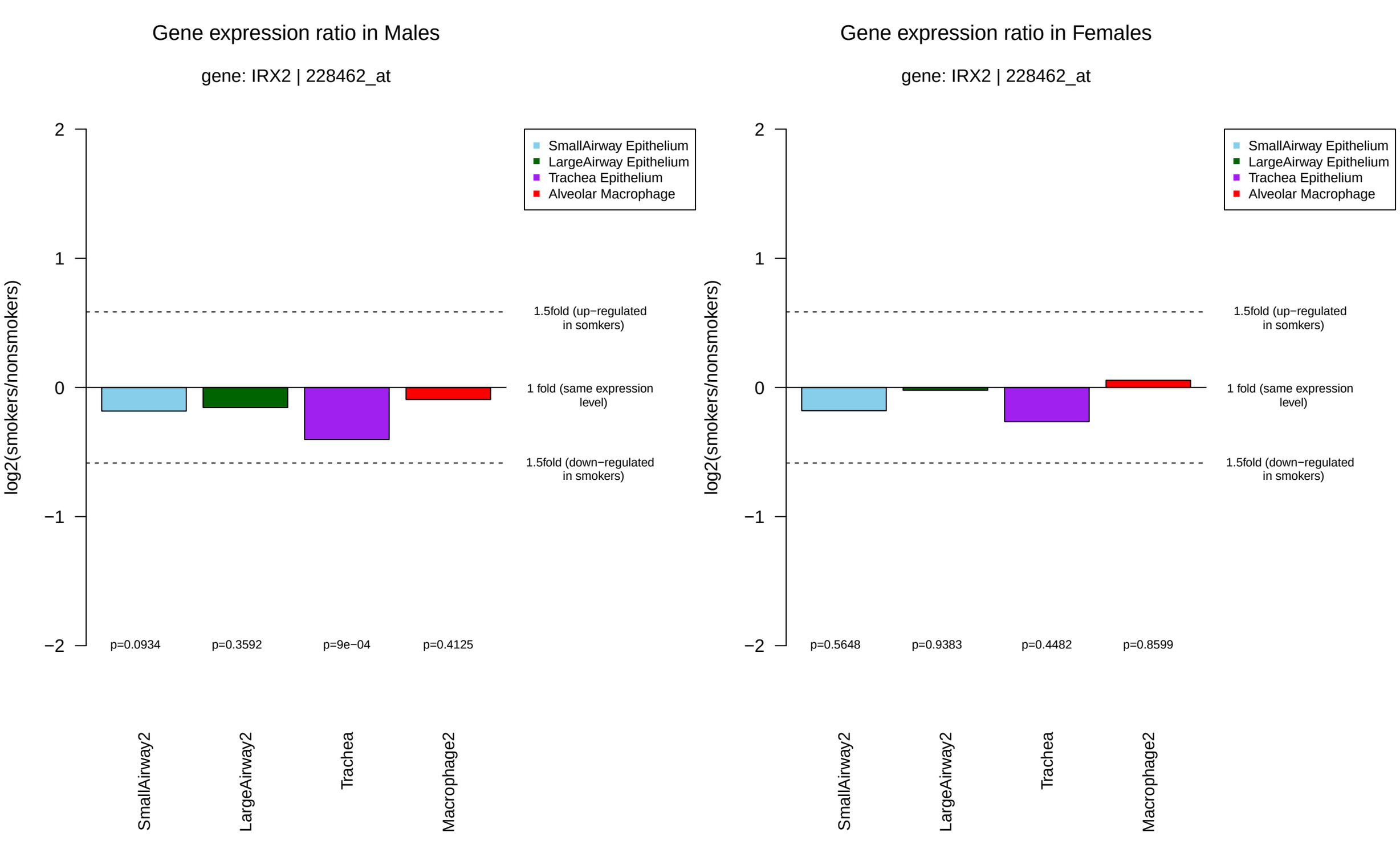
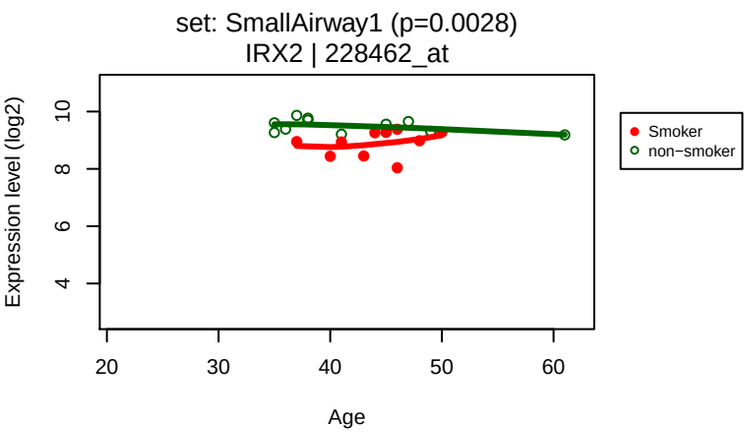
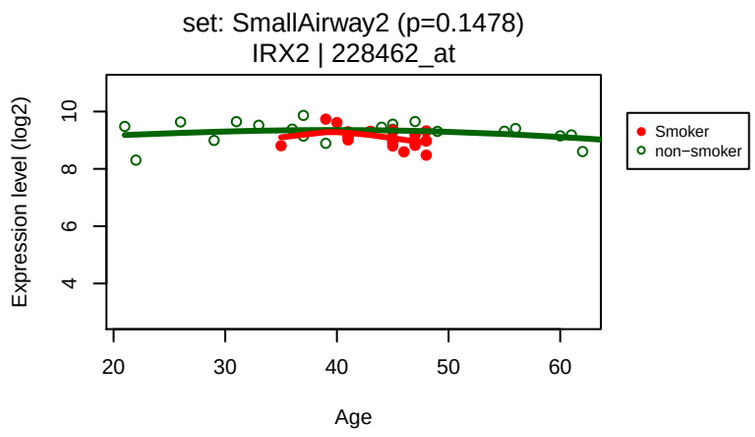
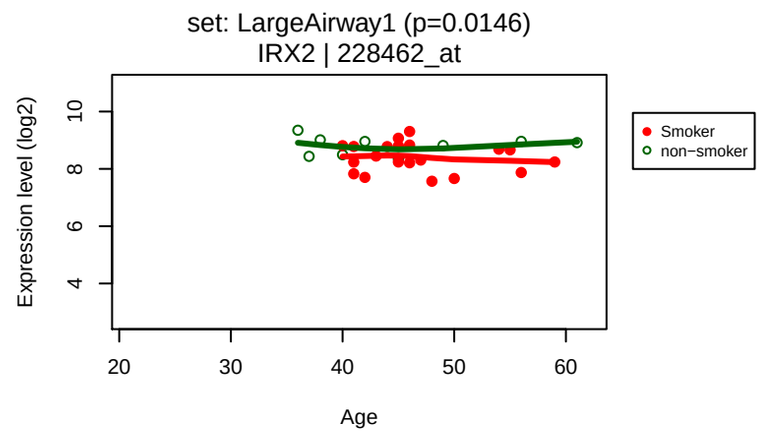

Gene expression of smokers and nonsmokers
gene: IRX2 | 228462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0028
p=0.1478
p=0.005
p=0.0146
p=0.5033
p=0.0011
p=0.5393
p=0.6752
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

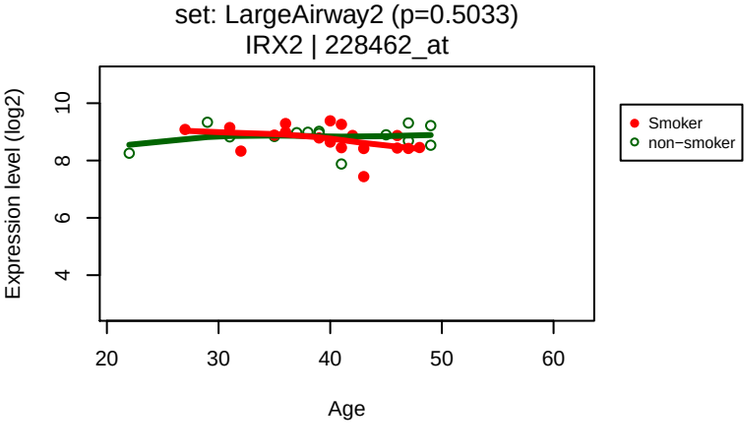

Gene expression ratio (smokers/nonsmokers)
gene: IRX2 | 228462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0028
p=0.1478
p=0.005
p=0.0146
p=0.5033
p=0.0011
p=0.5393
p=0.6752
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

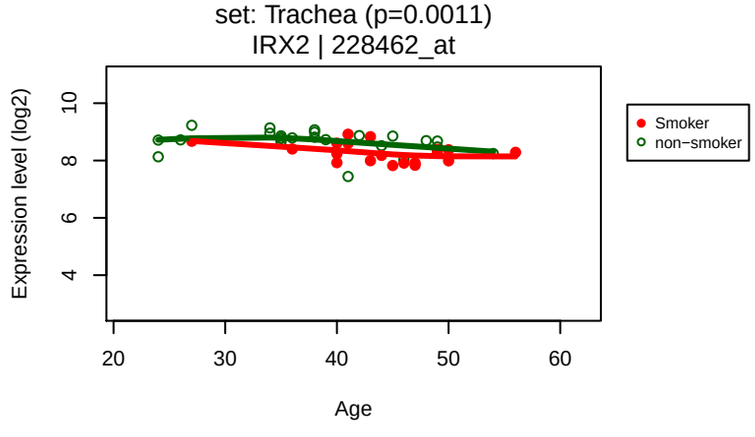

Gene expression ratio in Males
gene: IRX2 | 228462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0934
p=0.3592
p=9e−04
p=0.4125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRX2 | 228462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5648
p=0.9383
p=0.4482
p=0.8599
SmallAirway2
LargeAirway2
Trachea
Macrophage2

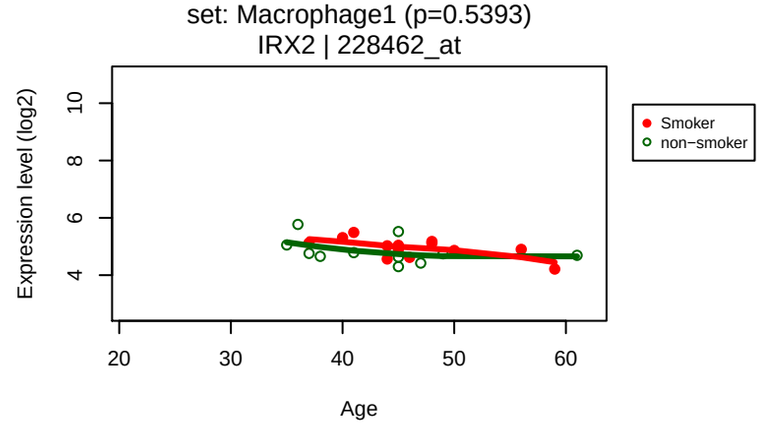

set: SmallAirway1 (p=0.0028)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker

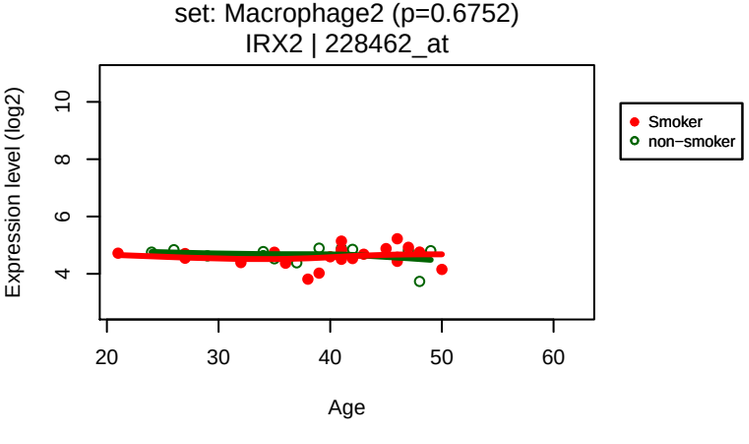

set: SmallAirway2 (p=0.1478)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker

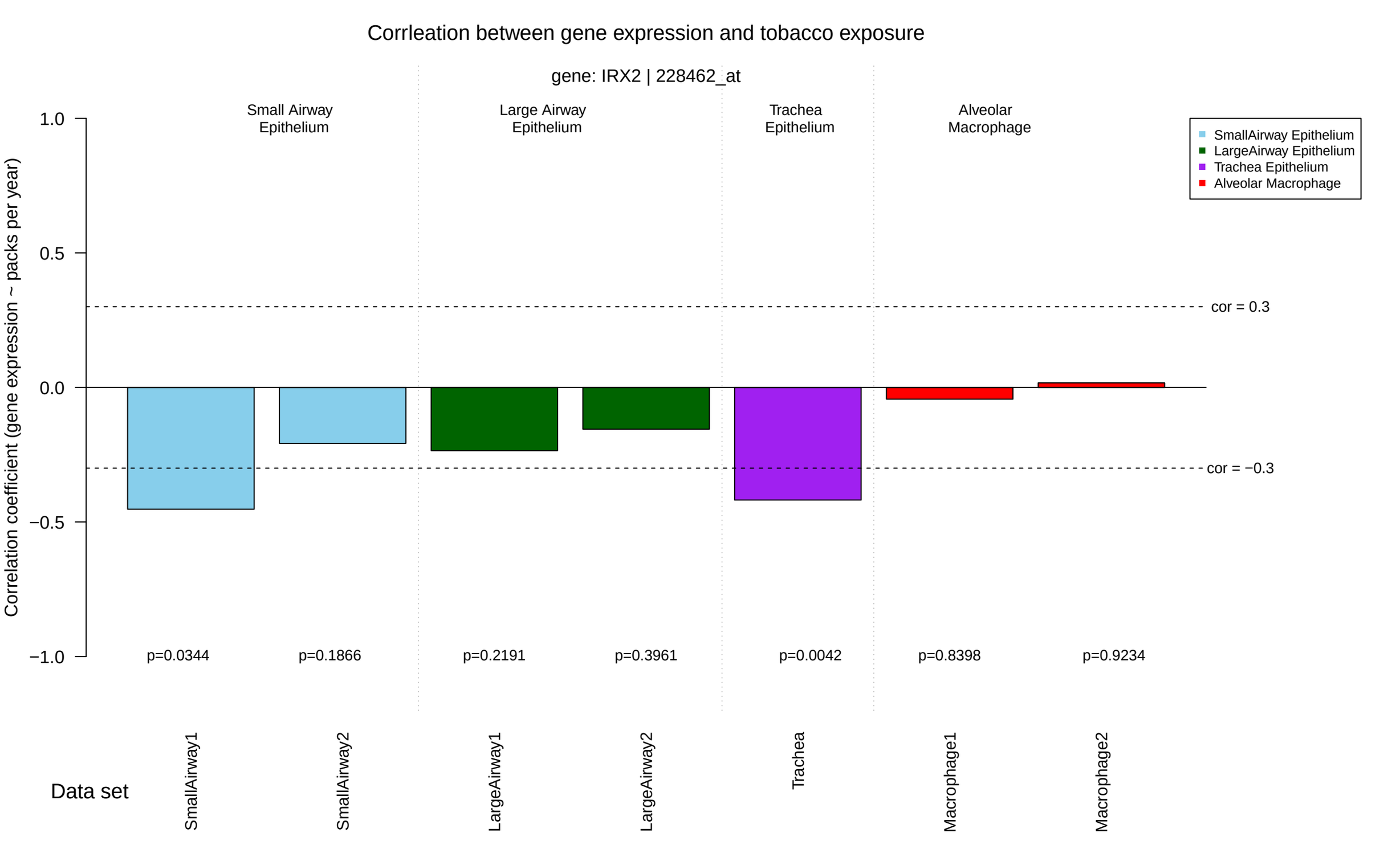

set: LargeAirway1 (p=0.0146)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5033)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0011)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5393)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6752)
IRX2 | 228462_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRX2 | 228462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0344
p=0.1866
p=0.2191
p=0.3961
p=0.0042
p=0.8398
p=0.9234
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2